

Direct density estimation using muon radiography

Cite as: J. Appl. Phys. 138, 014901 (2025); doi: 10.1063/5.0273292

Submitted: 29 March 2025 · Accepted: 12 June 2025 ·

Published Online: 1 July 2025

J. Peña-Rodríguez^{a)}

AFFILIATIONS

Fakultät für Mathematik und Naturwissenschaften, Bergische Universität Wuppertal, Wuppertal, Germany

Note: This paper is part of the Special Topic on Muography: Discoveries, Innovations, and Applications.^{a)}Author to whom correspondence should be addressed: penarodriguez@uni-wuppertal.de

ABSTRACT

Muography or muon radiography estimates the density distribution of natural or anthropic structures by measuring the traversing flux of atmospheric muons. Muography has been used to image volcanoes, glaciers, tunnels, line-shores, pyramids, and dams. The mass variation in the structure is deduced by the ratio between the open-sky muon flux and the target traversing flux, resulting in a relative measurement of the density. We present a novel method for directly measuring the average density of the target without prior knowledge or initial constraints. The methodology utilizes known muography variables, including open-sky and traversing muon flux, incident zenith angles, and muon path lengths. We validated the method with muography-simulated data from iron, aluminum, standard rock, and water phantoms, as well as real data from the Khufu pyramid, the Eiger Glacier, and the Canfranc Underground Laboratory.

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0273292>

I. INTRODUCTION

Muon radiography uses atmospheric muon flux to scan the inner density distribution of natural or human-made structures. Primary cosmic rays of extraterrestrial origin (~90% protons) interact continuously with the air nuclei, producing showers of secondary particles. The muon flux starts at the top of the Earth's atmosphere as a result of the decay of secondary particles such as kaons ($K^\pm \rightarrow \mu^\pm + \nu_\mu(\bar{\nu}_\mu)$ [~63.55%] and $K^\pm \rightarrow \pi^\pm + \pi^0$ [~20.66%]) and pions ($\pi^\pm \rightarrow \mu^\pm + \nu_\mu(\bar{\nu}_\mu)$ [~100%]). Muons have a high penetration power due to their small energy loss (~2 GeV in the whole atmosphere), relatively long lifetime (~2.2 μ s), and low cross section (electromagnetic and weak interactions). Muon interactions and the thickness of the Earth's atmosphere shape the muon flux at ground level. The overall angular distribution of the muons follows $\sim \cos^2 \theta$, where θ is the zenith angle of the incident muon up to 70°.¹

Muons interact weakly with matter, losing energy mainly by ionization or Bremsstrahlung. Muography uses muon penetration power to scan large natural or anthropic structures. Muography has been applied to monitor pyramids,^{2,3} volcanoes,^{4–10} nuclear reactors,¹¹ glaciers,^{12–14} tunnels,^{15,16} mines,^{17,18} and dams.^{19,20}

The muon flux attenuation after traversing the target can be modeled by

$$I = I_0 e^{-\int_L \mu dx}, \quad (1)$$

where I is the traversing muon flux, I_0 is the open sky muon flux, μ is the muon attenuation coefficient, and L is the muon path length.^{21,22} The attenuation coefficient depends on the muon energy, the interaction mechanisms, and the material properties. The transmittance T is

$$T = \frac{I}{I_0} = e^{-\int_L \mu dx}. \quad (2)$$

The muon attenuation coefficient is $\mu = \kappa \rho$, where κ is the mass attenuation coefficient in ($\text{cm}^2 \text{g}^{-1}$) and ρ is the density of the material. Then, we define the opacity by²³

$$\varrho = \int_L \rho dx = -\frac{1}{\kappa} \ln T. \quad (3)$$

Assuming an average density of the material along the muon path,

$$\varrho = \bar{\rho} L. \quad (4)$$

Muography estimates relative changes in the target density, and in some cases, a known density sample of the scanned target is used as a reference to obtain the density map.²⁴ Several methods reconstruct three-dimensional density distributions in muography:

linearized density length-based inversion,^{25–27} Bayesian joint inversion,^{28–30} filtered back projection,³¹ and algebraic reconstruction techniques.² All the methods mentioned have basic requirements like a minimum number of two-dimensional muography maps, where the density distribution is estimated based on prior knowledge from density measurements, simulations, a combination with gravity or electric resistivity data, or an initial density guess. In this work, we propose a novel method for estimating the average density along a given muon path directly from the muon flux without prior knowledge of the target composition.

II. DENSITY ESTIMATION PRINCIPLE

The open-sky muon flux for a given zenith angle θ is

$$I_0(\theta) = \int_0^\infty \Phi(E, \theta) dE \quad [\text{cm}^{-2} \text{sr}^{-1} \text{s}^{-1}]. \quad (5)$$

The muon flux passing through a given structure is the integration of the differential muon flux $\Phi(E, \theta)$ from the minimum crossing energy $E_{\min}$ to the infinite,

$$I(\theta, \varphi) = \int_{E_{\min}}^\infty \Phi(E, \theta) dE. \quad (6)$$

The traversing flux also depends on the azimuth angle φ because the target material budget varies for all (θ, φ) combinations. The missed muon flux can be obtained by subtraction of the traversing flux from the open-sky flux,

$$I_{\text{miss}}(\theta, \varphi) = \int_0^{E_{\min}} \Phi(E, \theta) dE = I_0(\theta) - I(\theta, \varphi). \quad (7)$$

We deduce $E_{\min}$ by minimizing the square of the difference between the function $I_0(E_{th}, \theta)$ and the measured traversing flux I .

The characteristic $I_0(E_{th}, \theta)$ is obtained by integrating the muon spectrum from a threshold energy (E_{th}) to the infinite, as shown in the left-hand side of Fig. 1. The minimizing function is given by

$$E_{\min} = \underset{E_{th}}{\operatorname{argmin}} [I_0(E_{th}, \theta) - I(\theta, \varphi)]^2. \quad (8)$$

Having $E_{\min}$ and the muon path length L , the material density is estimated by^{32,33}

$$\rho = \frac{300 \times 10^3}{L} \ln(1.5E + 1), \quad (9)$$

where ρ in g cm^{-3} , L in cm, and E in TeV. The model parameters were obtained by fitting the minimum muon energy necessary for traversing a distance L in different materials of density ρ . There exists only one average density ρ along the muon path L , which attenuates the incident muon flux $I_0(\theta)$ down to the observed flux crossing the target $I(\theta)$ as shown in Fig. 1 (right).

In a real implementation, we have to take into account the muon detector working principle. The detector records muon counting N for different trajectories (θ, φ) . The detected muon flux is estimated by

$$I(\theta, \varphi) = \frac{\eta(\theta, \varphi) N(\theta, \varphi)}{\mathcal{T}(\theta, \varphi) T}, \quad (10)$$

where $\mathcal{T}$ is the detector acceptance, T is the observation time, and η is the detector efficiency. The errors of the proposed methodology come mainly from the muon flux estimation, the muon path length error, and the detector parameters.

III. MUON FLUX MODEL

We simulated the muon differential flux using the MUYSC code³⁴ running the Reyna–Bugaev model.³⁵ In the Reyna–Bugaev

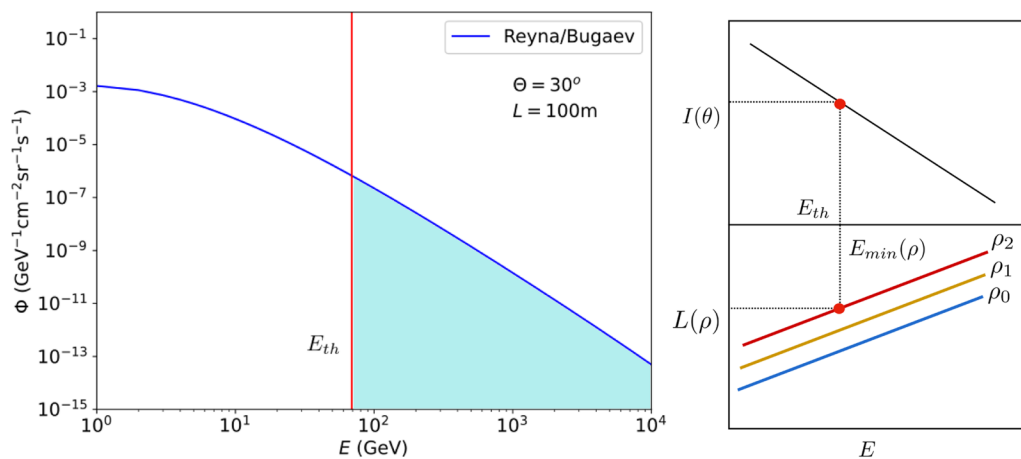


FIG. 1. Integration of the differential muon flux. The shadow area represents the traversing muon flux along 100 m of standard rock at 30° zenith (left). Direct density estimation principle (right). Minimum muon energy $E_{\min}$ of the measured muon flux $I(\theta)$ necessary for crossing the path length L along a target of density ρ_2 .

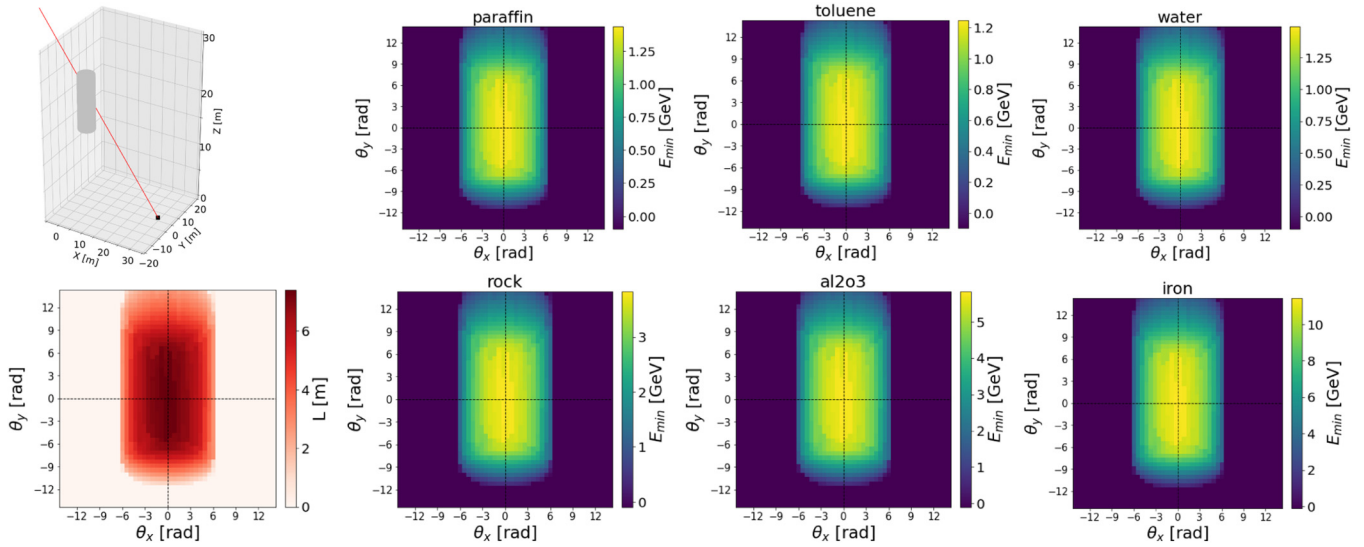


FIG. 2. Simulated cylinder phantom (top-left). Path length histogram of the cylinder with dimensions 10 m height and 6 m diameter (bottom-left). Minimum muon energy required for traversing the phantom depending on the path length and the material (paraffin, toluene, water, standard rock, aluminum oxide, and iron).

model, the muon flux is

$$\Phi_R(p, \theta) = A_R \hat{p}^{-(a_3 y^3 + a_2 y^2 + a_1 y + a_0)} \cos^3 \theta, \quad (11)$$

where p is the muon momentum in (GeV/c), θ is the incidence zenith angle, $y = \log_{10} \hat{p}$, and $\hat{p} = p \cos \theta$. The fitting parameters are $A_R = 0.00253$, $a_0 = 0.2455$, $a_1 = 1.288$, $a_2 = -0.2555$, and $a_3 = 0.0209$. The Reyna-Bugaev model shows the reliability for muon energies up to $\sim 10^3$ GeV and zenith angles up to 75° .³⁴

The atmospheric muon flux is dominated by low-energy muons and is affected by phenomena such as solar or geomagnetic modulation, atmospheric temperature/pressure, and altitude.²³

The solar modulation of the muon flux is caused by the Sun activity cycles, causing a variation of up to 1% (10%) at 10 GeV (1 GeV) between the solar maximum and minimum. Those cycles have a period of around 11 years, representing a minor issue for muography recording times (months). Atmospheric changes in pressure or temperature affect the muon flux, showing daily cycles; however, new muon telescopes implement flux corrections depending on the atmospheric temperature and pressure recorded during muon flux data collection.¹⁰ The geomagnetic latitude modulates the muon flux around 5% at 0° latitude. This modulation decreases with latitude.³⁶ Muography estimates the object density by measuring relative muon flux changes (target/open sky) at a fixed geographical position,

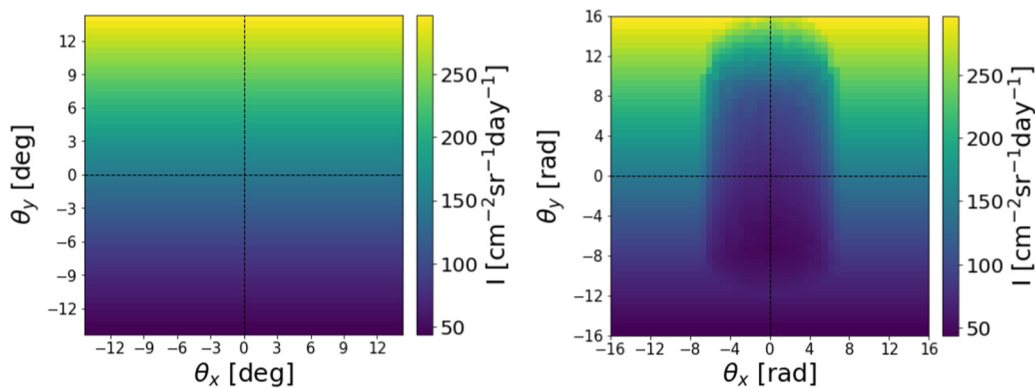


FIG. 3. Open-sky (left) and traversing muon flux across the standard rock phantom (right).

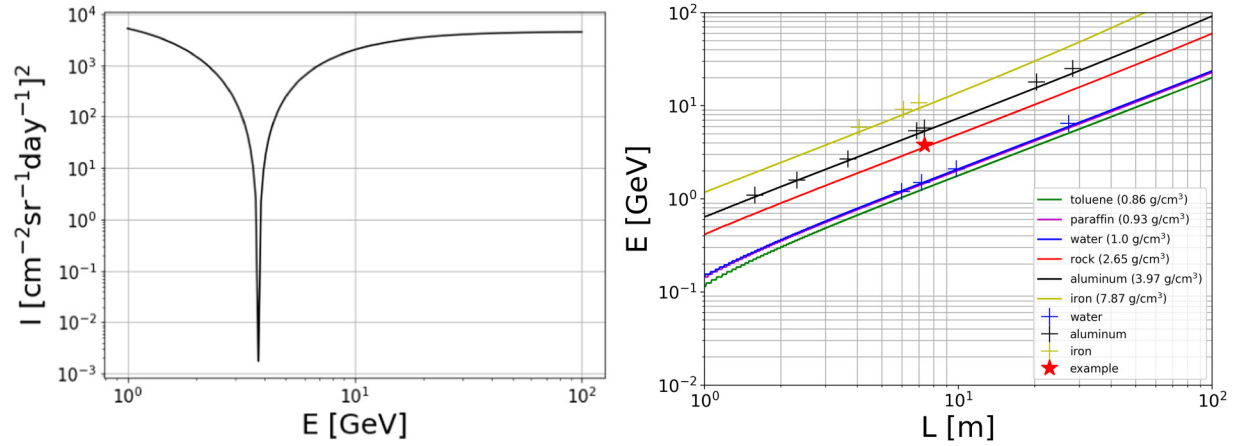


FIG. 4. Energy required [Eq. (13)] for traversing 7.36 m of standard rock at 57° zenith (left). Muon energy vs path length for different phantoms of toluene (green), paraffin (cyan), water (blue), standard rock (red), aluminum (black), and iron (yellow) (right). Crosses represent the model prediction. The red star shows the example described in the text (57° zenith, $67.34 \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$ traversing muon flux, $143.4 \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$ open-sky muon flux, and 7.36 m path length).

meaning that density estimation is not affected by the latitude; however, the absolute muon flux and the recording time are.

The MUYSC code performs altitude corrections,

$$\frac{\phi(h)}{\phi(h=0)} = e^{h/h_0}, \quad (12)$$

where h is the altitude of the observation place (in m), $h_0 = 4900 \text{ m} + 750 \text{ m} \frac{p}{\text{GeV}/c}$, and p is the muon momentum.³⁷

IV. METHOD VALIDATION

A. Simulations

We performed muon flux simulations for toluene (0.86 g cm^{-3}), paraffin (0.93 g cm^{-3}), water (1.0 g cm^{-3}), standard rock (2.65 g cm^{-3}), aluminum oxide (3.97 g cm^{-3}), and iron (7.87 g cm^{-3}) phantoms, as shown in Fig. 2. The phantom is a cylinder 10 m height and 6 m diameter. The observation point is located at 30 m from the cylinder center, with an elevation angle of 33° , and an aperture of $-14.3^\circ \leq \theta_{x,y} \leq 14.3^\circ$. The muon flux was

simulated at sea level, where the telescope $\theta_{x,y} = 0$ is equivalent to 57° zenith.

The muon energy loss in the phantom material was simulated with a parametric model implemented in the MUYSC code^{23,34} and based on the Groom data tables.³³ Figure 3 shows the traversing muon flux for the standard rock phantom. At $\theta_{x,y} = 0$ (57° zenith), the traversing muon flux is $67.34 \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$ with a path length of 7.36 m, while the open-sky muon flux is $143.4 \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$. The proposed method found a threshold muon energy of 3.77 GeV, as shown in the left-hand side of Fig. 4.

The simulated data fit accurately with the expected E vs L curves for the different materials as shown in the right-hand side of Fig. 4; taking into account that we know exactly the input parameters (the open-sky and traversing flux, the path length, and the incident zenith angle), the method operates with high accuracy. The estimated density values of the simulated phantoms are summarized in Table I.

The estimated density has a maximum error of about 10% for iron, $\sim 7.5\%$ for standard rock and aluminum oxide, and $<5\%$ for water, paraffin, and toluene.

B. Data

The density estimation method was tested with real muography data from the Khufu pyramid (Egypt),² the Eiger glacier (Switzerland, European Alps),¹⁴ and the Canfranc Underground Laboratory.³⁸ In those cases of study, we lack open-sky data, but we used the Reyna-Bugaev model for estimating it. The detected open-sky muon flux can vary slightly from the modeled flux depending on several factors, i.e., the detector calibration. We performed a correction of the modeled open-sky flux as follows:

$$E_{\min} = \underset{E_{th}}{\operatorname{argmin}} [\alpha I_0(E_{th}, \theta) - I(\theta, \varphi)]^2, \quad (13)$$

TABLE I. Density estimation of the simulated phantom.

Material	Estimated density (g cm^{-3})	Expected density (g cm^{-3})	Estimated/expected ratio
Toluene	0.86	0.86	1.00
Paraffin	0.97	0.93	1.05
Water	1.01	1.00	1.01
Standard rock	2.45	2.65	0.92
Aluminum oxide	3.67	3.97	0.92
Iron	7.79	7.87	0.99

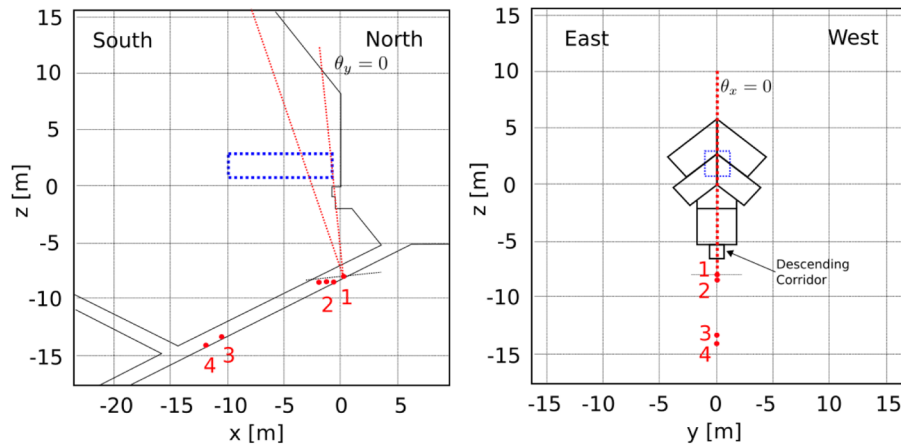


FIG. 5. Location of the detectors EM1, EM2N, EM2C, EM2S, EM3, and EM4 inside the Khufu pyramid.

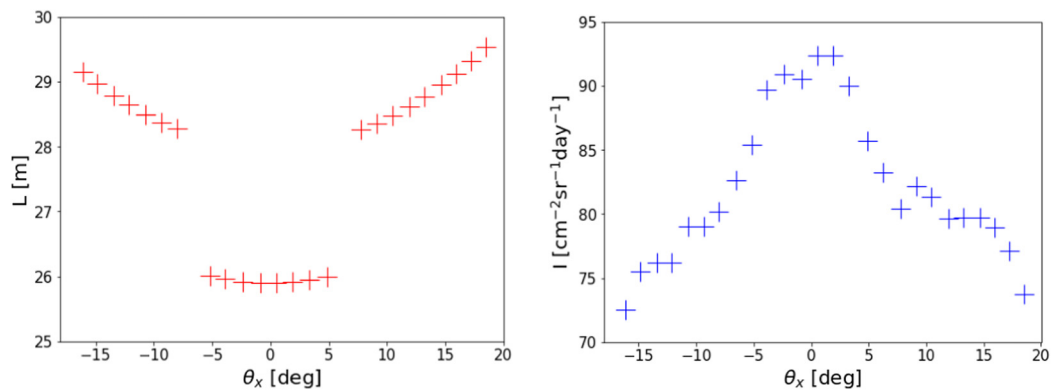


FIG. 6. Estimated muon path length in rock (left) and measured muon flux at the EM1 observation point (right).

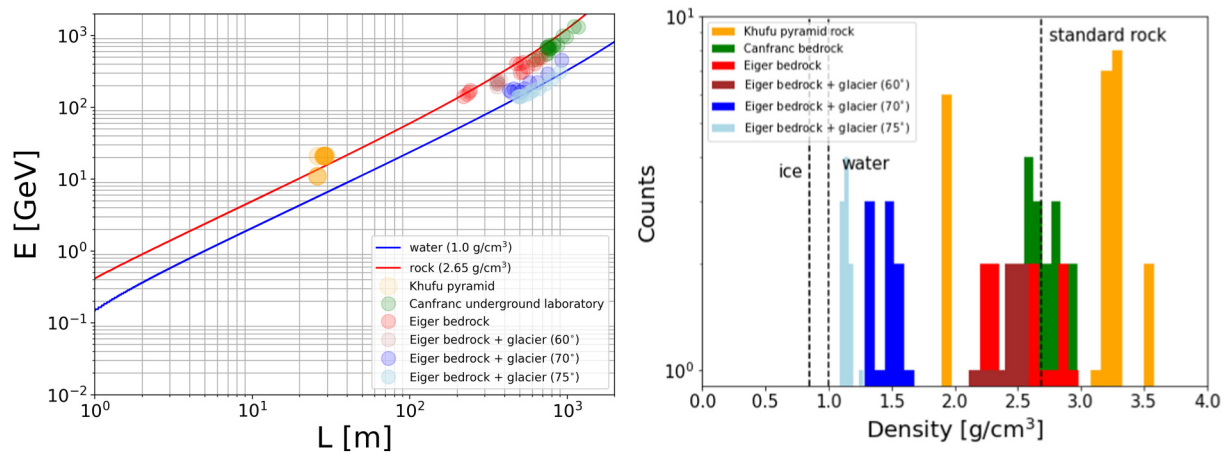


FIG. 7. Method validation with data from the Khufu pyramid rock (yellow dots), the Eiger glacier (blue/brown dots), and the Canfranc underground laboratory (green dots). The expected density of standard rock and water is represented by the red and blue solid lines, respectively (left). Density distribution of the Khufu pyramid rock, the Eiger glacier, and the bedrock of the Canfranc underground laboratory (right).

TABLE II. Density estimation.

Material	Estimated density (g cm ⁻³)	Expected density (g cm ⁻³)	Estimated/expected ratio
Khufu pyramid rock	2.93	2.65	1.10
Canfranc laboratory bedrock	2.69	2.73	0.99
Eiger bedrock	2.56	2.68	0.95
Eiger bedrock + glacier (60°)	2.37	2.56 ^a	0.92
Eiger bedrock + glacier (70°)	1.46	2.56	0.57
Eiger bedrock + glacier (75°)	1.15	2.56	0.44

^aAssuming as reference the estimated bedrock density.

where α is a scaling factor defined as the ratio measured/modeled open-sky flux at a given zenith angle in the observation site.

1. Khufu pyramid

The Great Pyramid is the largest of the three main pyramids in Egypt with a height of 146 m and served as the tomb of Khufu. The Nagoya University muography group installed six muon detectors (EM1, EM2N, EM2C, EM2S, EM3, and EM4) made of nuclear emulsion films in the Khufu pyramid to look for hidden chambers. The detectors had an angular range of $|\tan \theta_{x,y}| \leq 1.0$ (45°). We used the flux recorded by the EM1 detector because of its small uncertainty. EM1 took data during 172 days recording 9.48×10^7 tracks.

Despite the fact that the main goal of the campaign was to explore hidden voids in the pyramid structure, our aim is to estimate the density of the rock around the detector EM1. Figure 5 shows the configuration of the emulsion detectors inside the Khufu pyramid and the location of the hidden chamber. Figure 6 (left) and 6 (right) show the muon path and the measured muon flux, respectively.

The EM1 detector is located almost perpendicularly to the zenith 0 covering the same zenithal aperture for θ_x and θ_y . In general, the muon flux varies with the zenith angle, but remains constant with the azimuthal angle. EM1 measures the muon flux in the range $-26.5^\circ \leq \theta_x < 26.5^\circ$ in the region of $-14^\circ \leq \theta_y < -12^\circ$.

The dimensions of the chamber obtained by the monitoring campaign establish 2.18 ± 0.17 m height, 2.02 ± 0.06 m width, and 9.06 ± 0.07 m length. The EM1 point is located ~ 7.7 m below the base of the chamber and the slope of the external face of the pyramid is $\sim 51^\circ$. We calculated the path length of the muons that crossed the pyramid rock using the structural data described above as shown on the left-hand side of Fig. 6. The measured muon flux shows an enhancement in the area ($|\theta_x| \lesssim 5^\circ$) where the discovered chamber is located, as shown in the right-hand side of Fig. 6.

The calibration parameter $\alpha = 1.94$ was calculated taking into account that the open-sky muon flux at the zenith 0° is $\sim 1.2 \times 10^{-3} \text{ cm}^{-2} \text{ sr}^{-1} \text{ s}^{-1}$ according to the observation point G1 (Alhazen) of a previous expedition.⁵

The minimum muon energy necessary, to cross the different muon paths along the pyramid rock, is overlaid on the standard rock curve in the curve E vs L , as shown in the left of Fig. 7. The average rock density of the pyramid was estimated to be 2.93 g cm^{-3} , 10% higher than the reported, as shown in Table II.

2. Canfranc underground laboratory

The Canfranc Underground Laboratory is located under the Mount Tobazo (Aragonese Pyrenees, Spain) providing a perfect shielding for research in neutrino and dark matter physics. In Canfranc, there is a dedicated muon monitor composed of three layers of scintillation modules with a detection area of 1 m^2 and covering all azimuth angles and zenith angles up to 80° .³⁸

The muon detector was installed in two different places: LAB2400 and LAB2500. The path length traversed by muons arriving from a given direction was determined using topographic data

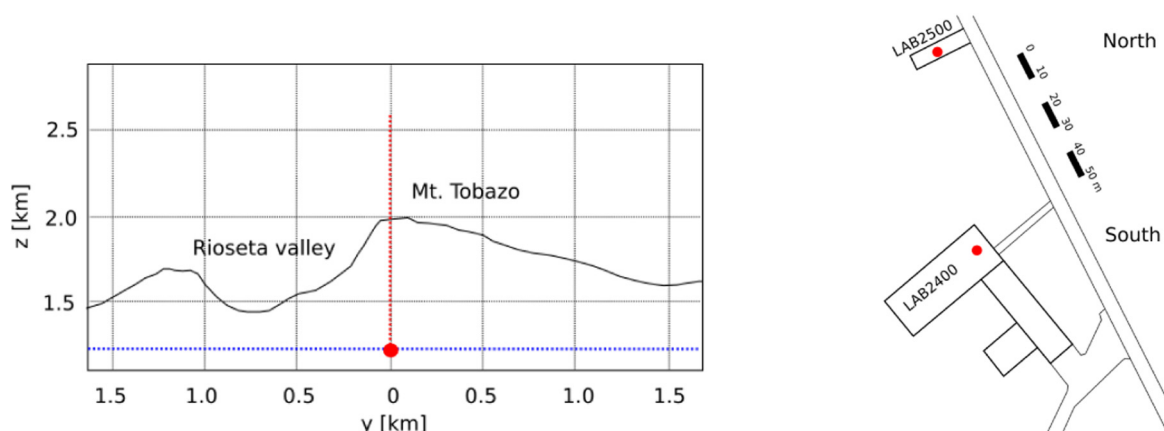


FIG. 8. Location of the Canfranc underground laboratory at the Mount Tobazo (left). Location of the muon detector at LAB2400 (1204.48 m a.s.l.) and LAB2500 (1206.47 m a.s.l.) (right).

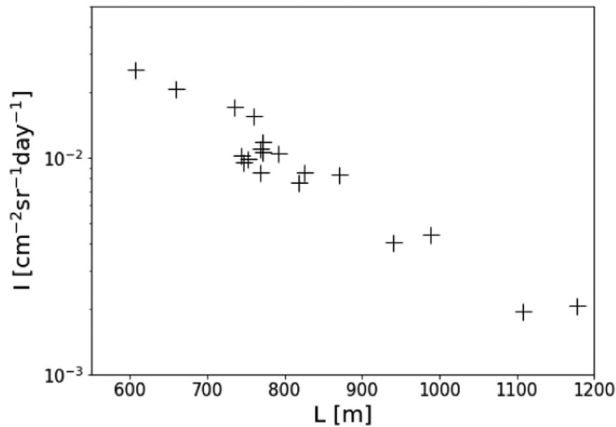


FIG. 9. Measured muon flux depending on the muon path length at 40° zenith at LAB2400 and LAB2500.

from the Advanced Land Observing Satellite (ALOS) (April 2018 release). Figure 8 shows the location and distribution of the underground laboratory of Canfranc.

The maximum muon flux ($25.5 \times 10^{-3} \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$) was observed in the direction of the Rioseta valley (40° zenith and 150° azimuth). Figure 9 shows the muon flux depending on the path length recorded from LAB2400 and LAB2500 inside the underground facility. The estimated density of the bedrock by the authors was 2.73 g cm^{-3} . We estimated a bedrock density of 2.69 g cm^{-3} , as shown in Table II.

3. Eiger glacier

In 2019, muon radiography was used on the flank of Mt. Eiger (Switzerland, European Alps) to reconstruct the shape of the bedrock and the Eiger glacier. Particle detectors made of nuclear emulsion films were installed in three different places: Eismeer

station (ES) 3159.9 m, tunnel site A (TA) 3186.4 m, and tunnel site B (TB) 3215.8 m, as shown in the left-hand side of Fig. 10.¹⁴ The bulk density of the bedrock was estimated from 16 rock samples and the mean value of 2.68 g cm^{-3} . They determined the dimensions of the glacier in terms of the ice–bedrock proportion along the muon trajectories from each detector, and then, the results were combined to get a three-dimensional representation. Figure 10 (right) shows the cross section of the Eiger glacier from the point of view of the TB detector.

The muon flux that traverses the glacier increases drastically in comparison with that of crossing pure bedrock, as shown in Fig. 11. The solid circles describe the muon flux detected by the TA station that only traverses the bedrock. The crosses show the muon flux detected by the TB station that traverses the glacier/bedrock.

We estimated the bulk density of the Eiger bedrock and ice from the measured muon flux shown in Fig. 11. We used the TA-measured muon flux ($0.846 \text{ cm}^{-2} \text{ sr}^{-1} \text{ day}^{-1}$) crossing 200 m bedrock at 45° zenith for calibrating our model ($\alpha \sim 3.6$). The calibration is necessary because we ignored the measured open-sky muon flux at the observation place.

Eiger Glacier data were split into four groups: bedrock, bedrock/ice at 60° zenith, bedrock/ice at 70° zenith, and bedrock/ice at 75° zenith. We estimated an average density of about 2.56 g cm^{-3} for the bedrock data. Then, the estimated density decreases as the observation zenith angle increases, 2.37 g cm^{-3} for 60° , 1.46 g cm^{-3} for 70° , and 1.15 g cm^{-3} for 75° . This tendency indicates that the muon path length is made up of a portion of bedrock and a portion of ice. The authors proposed a model from which we can estimate the bedrock and glacier dimensions along the path length,

$$\langle \rho \rangle = x\rho_{\text{rock}} + (1-x)\rho_{\text{ice}}, \quad (14)$$

where x is the bedrock portion. Then, $L_{\text{rock}} = Lx$ and $L_{\text{ice}} = L(1-x)$. Table II summarizes the estimated glacier/bedrock densities.

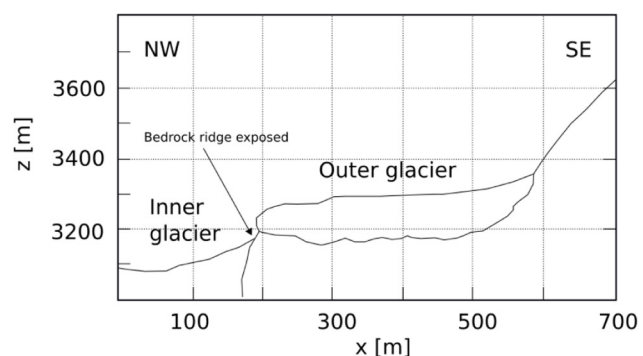
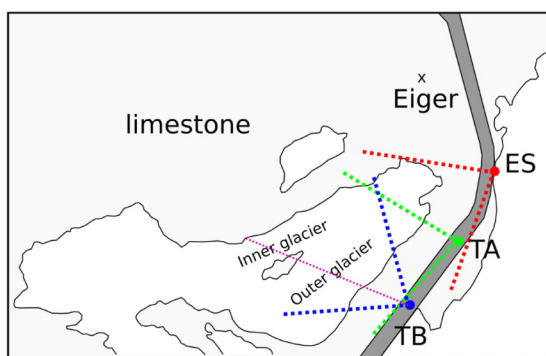


FIG. 10. Top view of the Eiger glacier. Location of the monitoring stations Eismeer station (ES) 3159.9 m, tunnel site A (TA) 3186.4 m, and tunnel site B (TB) 3215.8 m (left). Cross section of the Eiger glacier from tunnel site B.

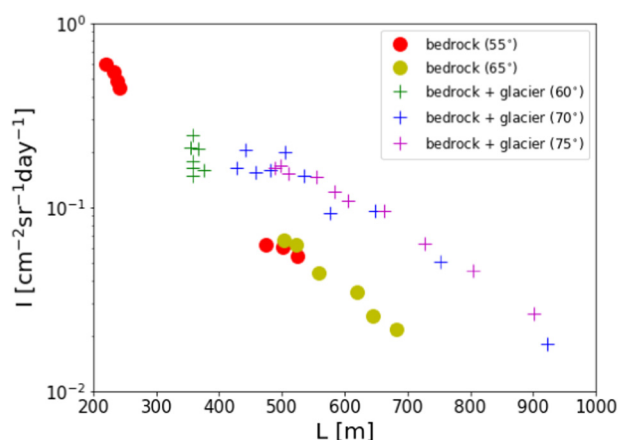


FIG. 11. Muon flux crossing the Eiger glacier. The solid circles indicate the muon flux that traverses only the bedrock (TA station), while the crosses show the muon flux crossing the glacier and the bedrock (TB station).

V. CONCLUSIONS

In this work, we presented a methodology for estimating the average density of the bulk material of a target using muography. Unlike other techniques, our method does not use prior knowledge (simulations/measurements) or initial constraints of the material density. We estimated the average density of six different simulated phantoms of toluene, paraffin, water, standard rock, aluminum oxide, and iron. The prediction error was below 10%. Furthermore, we performed a validation with real muography data from three independent campaigns, the Khufu pyramid, the Canfranc underground laboratory bedrock, and the Eiger glacier. The model predicted a density of 2.93 g cm^{-3} for the pyramid rock, 2.69 g cm^{-3} for the Canfranc laboratory bedrock and 2.56 g cm^{-3} for the bedrock of the Eiger glacier with a prediction error of $<10\%$. This approach also determined a density reduction in the Eiger glacier where an interface ice/bedrock is expected, helping to estimate the dimensions of the glacier as well as the bedrock formation.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

J. Peña-Rodríguez: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The energy loss data used for the phantom simulations are reported by Groom.³³ The MUYSC code for simulating the muon flux is available in GitHub at <https://github.com/Jhosgun/MUYSC>, Ref. 34.

REFERENCES

- ¹S. Cecchini and M. Spurio, "Atmospheric muons: Experimental aspects," *Geosci. Instrum. Methods Data Syst.* **1**, 185–196 (2012).
- ²S. Procureur *et al.*, "Precise characterization of a corridor-shaped structure in Khufu's pyramid by observation of cosmic-ray muons," *Nat. Commun.* **14**, 1144 (2023).
- ³K. Morishima *et al.*, "Discovery of a big void in Khufu's pyramid by observation of cosmic-ray muons," *Nature* **552**, 386–390 (2017).
- ⁴V. Tioukov *et al.*, "First muography of Stromboli volcano," *Sci. Rep.* **9**, 6695 (2019).
- ⁵L. Oláh *et al.*, "Plug formation imaged beneath the active craters of Sakurajima volcano with muography," *Geophys. Res. Lett.* **46**, 10417–10424, <https://doi.org/10.1029/2019gl084784> (2019).
- ⁶M. D'Errico *et al.*, "Muon radiography applied to volcanoes imaging: The MURAVES experiment at Mt. Vesuvius," *J. Instrum.* **15**, C03014 (2020).
- ⁷M. Rosas-Carbajal *et al.*, "Three-dimensional density structure of La Soufrière de Guadeloupe lava dome from simultaneous muon radiographies and gravity data," *Geophys. Res. Lett.* **44**, 6743–6751, <https://doi.org/10.1002/2017gl074285> (2017).
- ⁸H. K. M. Tanaka, "Japanese volcanoes visualized with muography," *Philos. Trans. R. Soc. A* **377**, 20180142 (2018).
- ⁹A. Vesga-Ramírez *et al.*, "Simulated annealing for volcano muography," *J. South Am. Earth Sci.* **109**, 103248 (2021).
- ¹⁰J. Peña-Rodríguez *et al.*, "Design and construction of MuTe: A hybrid muon telescope to study colombian volcanoes," *J. Instrum.* **15**, P09006 (2020).
- ¹¹S. Procureur *et al.*, "3D imaging of a nuclear reactor using muography measurements," *Sci. Adv.* **9**, eabq8431 (2023).
- ¹²R. Nishiyama *et al.*, "First measurement of ice-bedrock interface of alpine glaciers by cosmic muon radiography," *Geophys. Res. Lett.* **44**, 6244–6251, <https://doi.org/10.1002/2017GL073599> (2017).
- ¹³A. Ariga *et al.*, "A nuclear emulsion detector for the muon radiography of a glacier structure," *Instruments* **2**, 7 (2018).
- ¹⁴R. Nishiyama *et al.*, "Bedrock sculpting under an active alpine glacier revealed from cosmic-ray muon radiography," *Sci. Rep.* **9**, 6970 (2019).
- ¹⁵X. Mao *et al.*, "Muon radiography experiments on the subway overburden structure detection," *Nucl. Instrum. Methods Phys. Res., Sect. A* **1055**, 168391 (2023).
- ¹⁶L. F. Thompson *et al.*, "Muon tomography for railway tunnel imaging," *Phys. Rev. Res.* **2**, 023017 (2020).
- ¹⁷G. Liu *et al.*, "Deep investigation of muography in discovering geological structures in mineral exploration: A case study of Zaozigou gold mine," *Geophys. J. Int.* **237**, 588–603, <https://doi.org/10.1002/2017GL073599> (2024).
- ¹⁸D. Borselli, T. Beni, L. Bonechi, M. Bonghi, D. Brocchini, N. Casagli, R. Ciaranfi, L. Cimmino, V. Ciulli, R. D'Alessandro, A. Dini, C. Frosin, G. Gigli, S. Gonzi, S. Guideri, L. Lombardi, M. Nocentini, and G. Saracino, "Three-dimensional muon imaging of cavities inside the Temperino mine (Italy)," *Sci. Rep.* **12**, 22329 (2022).
- ¹⁹L. Oláh *et al.*, "Structural health monitoring of sabo check dams with cosmic-ray muography," *iScience* **26**, 108019 (2023).
- ²⁰I. Lázaro Roche, "A compact muon tracker for dynamic tomography of density based on a thin time projection chamber with micromegas readout," *Particles* **4**, 333–342 (2021).
- ²¹R. N. Altameemi, N. S. Abdul Hamid, W. M. A. Wan Mohd Kamil, and S. M. Saleh Ahmed, "Theoretical formula for the cosmic-ray muon attenuation coefficients of Pb, Cu, Zn, and Al," *J. Radiat. Res. Appl. Sci.* **13**, 721–729 (2020).
- ²²Z. Cao *et al.*, "Measurement of attenuation length of the muon content in extensive air showers from 0.3 to 30 PeV with LHAASO," *Phys. Rev. D* **110**, 103017 (2024).
- ²³N. Lesparre *et al.*, "Geophysical muon imaging: Feasibility and limits," *Geophys. J. Int.* **183**, 1348–1361, <https://doi.org/10.1111/j.1365-246X.2010.04790.x> (2010).
- ²⁴N. Lesparre *et al.*, "Density muon radiography of La Soufrière de Guadeloupe volcano: Comparison with geological, electrical resistivity and gravity data,"

Geophys. J. Int. **190**, 1008–1019, <https://doi.org/10.1111/j.1365-246X.2012.05546.x> (2012).

²⁵L. Balázs, G. Nyitrai, G. Surányi, G. Hamar, G. G. Barnaföldi, and D. Varga, “3D muographic inversion in the exploration of cavities and low-density fractured zones,” *Geophys. J. Int.* **236**, 700–710, <https://doi.org/10.1093/gji/ggad428> (2023).

²⁶D. Schouten and P. Ledru, “Muon tomography applied to a dense uranium deposit at the McArthur river mine,” *J. Geophys. Res.: Solid Earth* **123**, 8637–8652, <https://doi.org/10.1029/2018JB015626> (2018).

²⁷S. Nagahara, S. Miyamoto, K. Morishima, T. Nakano, M. Koyama, and Y. Suzuki, “Three-dimensional density tomography determined from multi-directional muography of the Omuroyama scoria cone, Higashi-Izu monogenetic volcano field, Japan,” *Bull. Volcanol.* **84**, 94 (2022).

²⁸R. Nishiyama, “Joint inversion of muography and gravity data for 3D density imaging of volcanoes,” in *Muography: Exploring Earth's Subsurface with Elementary Particles* (Wiley, 2022), Chap. 3.

²⁹K. Cosburn, M. Roy, and R. Nishiyama, “A machine learning approach to joint gravity and cosmic-ray muon inversion at Mt Usu, Japan,” *Geophys. J. Int.* **233**, 1081–1096, <https://doi.org/10.1093/gji/ggac497> (2022).

³⁰A. Barnoud, V. Cayol, V. Niess, C. Cârloganu, P. Lelièvre, P. Labazuy, and E. Le Ménèdeu, “Bayesian joint muographic and gravimetric inversion applied to volcanoes,” *Geophys. J. Int.* **218**, 2179–2194, <https://doi.org/10.1093/gji/ggz300> (2019).

³¹S. Miyamoto and S. Nagahara, “Tomographic imaging of volcano structures with cosmic-ray muons,” in *Muography: Exploring Earth's Subsurface with Elementary Particles* (Wiley, 2022), Chap. 2.

³²K. Nagamine *et al.*, “Method of probing inner-structure of geophysical substance with the horizontal cosmic-ray muons and possible application to volcanic eruption prediction,” *Nucl. Instrum. Methods Phys. Res., Sect. A* **356**, 585–595 (1995).

³³D. E. Groom *et al.*, “Muon stopping power and range tables 10 MeV–100 TeV,” *At. Data Nucl. Data Tables* **78**, 183–356 (2001).

³⁴J. Peña-Rodríguez *et al.*, “MUYSC: An end-to-end muography simulation toolbox,” *Geophys. Res. Lett.* **237**, 540–556, <https://doi.org/10.1093/gji/ggae064> (2024).

³⁵D. Reyna, “A Simple parameterization of the cosmic-ray muon momentum spectra at the surface as a function of zenith angle,” [arXiv:hep-ph/0604145](https://arxiv.org/abs/hep-ph/0604145) (2006).

³⁶C. Sarmiento-Cano, M. Suárez-Durán, R. Calderón-Ardila, A. Vásquez-Ramírez, A. Jaimes-Motta, L. A. Núñez, S. Dasso, I. Sidelnik, and H. Asorey, “The arti framework: Cosmic rays atmospheric background simulations,” *Eur. Phys. J. C* **82**, 1019 (2022).

³⁷T. Hebbeker and C. Timmermans, “A compilation of high energy atmospheric muon data at sea level,” *Astropart. Phys.* **18**, 107–127 (2002).

³⁸W. H. Trzaska *et al.*, “Cosmic-ray muon flux at Canfranc Underground Laboratory,” *Eur. Phys. J. C* **79**, 721 (2019).